



Production and MRP: Production Standard Cost

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the **Production Std. Cost** training topic.

- Although Production Standard Cost fields exist for both perpetual and non-perpetual inventory, we only cover the functionality for perpetual inventory in this course.
- In addition, in order to take this training, you first need to complete the **Production Process** and **By-products and Additional Quantity** training topics.

Objectives



At the end of this topic, you will be able to:

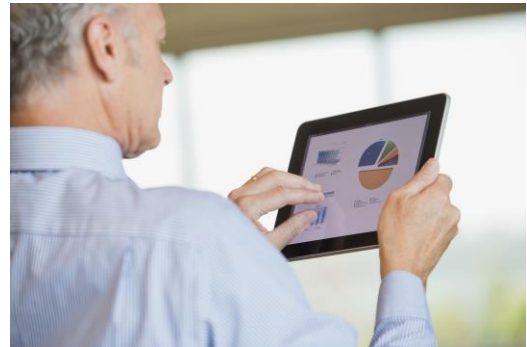
- Define a production standard cost for produced items
- Explain the total production standard cost calculation in the *Bill of Materials*
- Describe the production standard cost rollup and update procedures

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 - Explain the total production standard cost calculation in the Bill of Materials
 - Describe the production standard cost rollup and update procedures

Business Scenario



- OC WoodTrend produces custom wooden doors. They use the Production module in SAP Business One to record the production process.
- There was large fluctuation in the production cost of the doors over the last year and the accountant would like to have better cost prediction.
- OC WoodTrend decides to manage Production Standard cost in order to analyze and more accurately predict costs.
- In addition, she would like to set up the relevant cost accounts for the resources in the GL account determination.

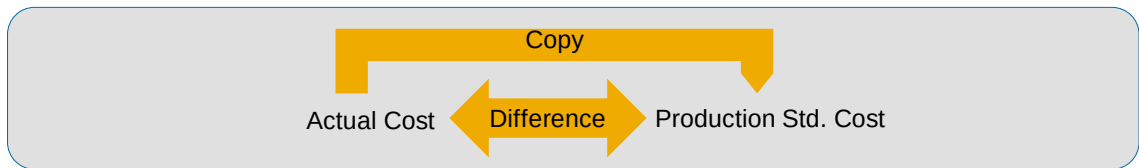


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- OC WoodTrend decides to manage Production Standard cost in order to analyze and more accurately predict costs.
- In addition, she would like to set up the relevant cost accounts for the resources in the GL account determination.

Concept



- Production std. cost is a budgeted production cost which can be compared to the actual production cost.
- *Item Master Data Production Std Cost* can be optionally set to the current valuation cost and can also be optionally rolled up through multiple levels of bill of materials.
- Bill of Materials window displays the *Production Std. Cost* data for item components, the cost data for resource components and the *Production Std Cost* data for the produced item.

Benefit: Facilitates budget to actual cost comparisons and production variance calculations.

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- The main idea behind this feature is to allow us to plan a budget for the production cost. Then we can compare the actual cost of a product to this budgeted cost. In addition we can copy the actual cost of the produced items in a BOM to their *Production Std. Cost* in an automatic procedure.
- When working in the *Bill of Material* and *Production Order* windows, we can see the *Production Std. Cost* information. In addition we can analyze the difference between the budgeted cost and the actual cost in the *Inventory Audit Trail* report.

Resource Std. Cost Definition

Resource Group

Resource Type: Machine
Unit of Measure Text:

Resource Std Cost	User-Definable Name	Default Std Cost
Resource Std Cost 1	Amortization	100
Resource Std Cost 2	Maintenance	50
Resource Std Cost 3	Overhead	10
Resource Std Cost 4	Resource Std Cost 4	
Resource Std Cost 5	Resource Std Cost 5	
Resource Std Cost 6	Resource Std Cost 6	
Resource Std Cost 7	Resource Std Cost 7	
Resource Std Cost 8	Resource Std Cost 8	
Resource Std Cost 9	Resource Std Cost 9	
Resource Std Cost 10	Resource Std Cost 10	

Resource Master Data

Resource Std Cost	Default Std Cost
Amortization	100.00
Maintenance	50.00
Overhead	0.00
Resource Std Cost 4	0.00
Resource Std Cost 5	0.00
Resource Std Cost 6	0.00
Resource Std Cost 7	0.00
Resource Std Cost 8	0.00
Resource Std Cost 9	0.00
Resource Std Cost 10	0.00
Total Std Resource Cost	150.00

G/L Account Determination

#	Type of Account	Account Code	Account Name
Std Cost Expense 1		510160	Resources Expenses
Std Cost Expense 2		510160	Resources Expenses
Std Cost Expense 3			
Std Cost Expense 4			
Std Cost Expense 5			
Std Cost Expense 6			
Std Cost Expense 7			
Std Cost Expense 8			
Std Cost Expense 9			
Std Cost Expense 10			

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- The cost of a resource is actually the std. cost that is defined in the *Resource Master data*. This cost is composed from a list of
- different std. cost elements that are copied from the *Resource Group*. For the lathe machine, the standard cost is comprised of 3 elements: amortization, maintenance and overhead.
- Each cost element is related to a certain account in the *GL Account Determination* window (when using standard G/L account determination). There, we can define up to ten resource expense accounts, corresponding to the ten cost components in the *Resource Master Data*.
- The cost element in each row in the *Resource Master Data* is linked by the element number to the cost expense fields in the G/L account definition. In OC WoodTrens, Resource Std. Cost **1** of the **Lathe** machine is named **Amortization** and refers to account **510160** as shown in the image. The same cost element may have a different name in another resource.
- In *Advanced GL Account Determination* it is possible to set rules that connect resource std. cost elements to different G/L accounts.

Item Std. Cost Definition

Item Master Data

Item No.	Manual	D00002	☒ Inventory Item
Description	Portable Hard Disk 2 TB		☒ Sales Item
Foreign Name			☒ Purchase Item
Item Type	Items		
Item Group	Items		
UoM Group	Manual		
Price List	Base Price		
		Bar Code	
		Unit Price	Primary Curre GBP 70.00

General Purchasing Data Sales Data Inventory Data Planning Data Production Data Properties Rep

☐ Phantom Item
Issue Method Backflush

BOM Type
No. of Item Components 0
No. of Resource Components 0
No. of Route Stages 0
Production Std Cost 200
☒ Include in Production Std Cost Rollup

Can be entered manually or populated during the *Production Std. Cost Rollup* procedure

- The *Production Std. Cost* of an item is defined manually in the *Item Master Data*.
- The *Production Std. Cost* value can also be copied on a batch basis from the actual product cost of the Item (including component items). This is done in the Rollup procedure. We can see in the image a checkbox indicating this item is included in the *Production Std. Cost Rollup*. We will explore this option later on in the training.

Production Std. Cost Calculation in the Bill of Materials

Bill of Materials

Product No. D00002 X Quantity 1 Warehouse 01
 Product Description Decorative Wooden Door Price List Base Price
 BOM Type Production Distr. Rule
 Project

Production Std Cost GBP 200.00
 Planned Average Production Size 10.00

#	Type	No.	Description	Quantity	UoM Name	Additional Quantity	Production Std Cost	Total Production Std Cost
1	Item	D00001	Wooden Door	1	Eac		GBP 100.00	GBP 100.00
2	Item	H00001	Handle	1			GBP 10.00	GBP 10.00
3	Resourc	R50001	Lathe Machine	2	Cycle		GBP 50.00	GBP 100.00
4	Text		Turn 180 degrees					
5	Resourc	R50001	Lathe Machine	1	Cycle		GBP 50.00	GBP 50.00
6	Resourc	RE0001	Machine Operator	2		1	GBP 30.00	GBP 63.00
7	Item							GBP 323.00

Product Price

OK Cancel

Production Std. Cost X Qty.

+

Additional Qty X $\frac{\text{Production Std. Cost}}{\text{Planned Avg. Prod. Size}}$

Total Production Std. Cost

- In the *Bill of Materials* we can see the *Production Std. Cost* that is taken from the *Item Master Data* as seen in the previous slide.
 - Beneath this field there is another field called: *Planned Average Production Size*. Here we enter the average quantity of produced items we usually produce in one production run.
 - In our example, we can say that on average, we produce **10 Decorative Wooden Doors** every time we go through a production process of **Decorative Wooden Doors**.
 - This value is relevant for the *Additional Quantity* we got to know in the **By-products and Additional Quantity** training.
- The *Production Std. Cost* of a produced item is equal to the cost of all its components plus the fixed cost of the additional quantity needed to produce one item. In our case, since we need **1 Machine Operator** hour, the fixed cost needed to produced **1 Decorative Wooden Doors** is **30** divided by a planned average production quantity of **10** multiplied by **1 Additional Quantity**. This means that the fixed cost (derived from the *Additional Quantity*) of producing **1 Decorative Wooden Door** is **3**.
- Therefore, the *Total Production Std. Cost* of the **Machine Operator** resource is **30**, times a quantity of **2** hours, plus **3** of fixed cost. This sums up to **63**.
 - At the bottom of the table we see the total production std. cost for the **Decorative Wooden Door**.
 - Note that this calculated total is different than the cost of **200**, that we see in the header and that is originated from the *Item Master Data*.
 - SAP Business one provides an automatic routine that updates the *Production Std. Cost* of the item according to the calculated cost in the bill of materials. Let us see it in the next slide.

Production Std. Cost Rollup

**Production → Production Std. Cost Management
→ Production Std. Cost Roll Up**

Inventory → Item Master Data → Production Tab

The image shows two SAP screenshots. On the left is the 'Production Std Cost Rollup - Selection Criteria' dialog box. It contains fields for 'Parent Item No.' with 'From' and 'To' sub-fields, an 'Item Group' dropdown menu set to 'All', and buttons for 'Item Properties', 'Ignore', 'OK', and 'Cancel'. On the right is the 'Production Std Cost' tab in the Item Master Data. It shows a checkbox labeled 'Include in Production Std Cost Rollup' which is checked. A yellow arrow points from this checkbox to the 'Item Properties' button in the dialog box. Another yellow arrow points from the 'Include in Production Std Cost Rollup' checkbox to a text box below the dialog.

Production Std Cost Rollup - Selection Criteria

Parent Item No. From To

Item Group All

Item Properties Ignore

OK Cancel

Production Std Cost

☒ Include in Production Std Cost Rollup

Relevant for produced items that are marked as included in the rollup procedure.

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- In **Production → Production Std. Cost Management → Production Std. Cost Roll Up**, we can choose which items to update.
- Once we choose **OK**, the *Production Std. Cost* in the *Item Master Data* is updated according to the calculated *Production Std. Cost* of the *Bill of Materials*.
- Before we run the roll up procedure, we need to make sure the items are marked as *Included in Production Std. Cost Rollup*.
- Note that the roll up procedure is calculated for all levels of the *Bill of Material*.

Production Std. Cost Update

Production → Production Std. Cost Management → Production Std. Cost Update

Production Std Cost Update - Selection Criteria

Update Production Standard Cost from Current Valuation Cost

Item No. From To

Item Group All

Item Properties Ignore

OK Cancel

Relevant for all items



- First perform a *Production Std. Cost Update*
- Then the Std. Cost Rollup procedure

- The rollup procedure updated the Production Std. Cost of produced items according to their components' Production Std. Cost.
- In the *Production Std. Cost Update* window we can update the items' *Production Std. Cost*, according to their actual cost.
- This procedure works similarly to the rollup procedure. In the *Production Std. Cost Update* window, choose the relevant items and choose the *Ok* button. The update is done instantly.
- Note that in the *Selection Criteria window* you can choose any item, even if it is not a produced item nor a component.
- In a manufacturing company, it is recommended to first perform a *Production Std. Cost Update* and then a rollup procedure. This way we update the actual cost of the components and then roll it up to the *Production Std. Cost* of the produced item.
- Note that this feature does not apply to non-perpetual inventory managed companies.

Key Points



Key points from this topic:

- The *Production Std. Cost* feature allows us to plan a budget for the production cost.
- The Production Std. cost can be entered manually or copied via the rollups procedure.
- The *Production Std. Cost Rollup procedure updates the Production Std. Cost* in the *Item Master Data of the produced items* according to the calculated *Production Std. Cost* in the *Bill of Materials*.
- In the *Production Std. Cost Update* window we can update the items' *Production Std. Cost*, according to their actual cost. This feature is relevant for all items.

- Here are some key points to take away from this session:
 - The *Production Std. Cost* feature allows us to plan a budget for the production cost.
 - The Production Std. cost can be entered manually or copied via the rollups procedure.
 - The *Production Std. Cost Rollup procedure updates the Production Std. Cost* in the *Item Master Data of the produced items* according to the calculated *Production Std. Cost* in the *Bill of Materials*.
 - In the *Production Std. Cost Update* window we can update the items' *Production Std. Cost*, according to their actual cost. This feature is relevant for all items

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